

Deepak Mallampati

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Kent, OH - US-based, F-1 OPT (open to sponsorship)

PROFESSIONAL SUMMARY

Applied AI Engineer with production Claude integration, healthcare RAG depth, and nonprofit deployment experience. Shipped Manga Lens to the Chrome Web Store with **Claude Sonnet** as one of four vision providers - production Claude integration in user-facing software. Built healthcare RAG with **~35% retrieval precision gain** and **>90% grounding alignment** in HIPAA-conscious environment at Progress Solutions. Built nonprofit applied AI at Suvidha Foundation - transformer summarization + document Q&A RAG for non-technical users (60-70% manual review time reduction across 5,000+ sessions). 5-stage multi-agent claims pipeline with **schema-validated JSON contracts** (MCP-adjacent) prevents cascading hallucinations with audit-ready traces. Scrappy builder. Master's in CS (Kent State). US-based, F-1 OPT, open to sponsorship.

CORE COMPETENCIES

Claude API Integration • Beneficial Deployments • Healthcare AI • Nonprofit AI Delivery • MCP-Adjacent Multi-Agent • Agent Skills & Sub-Agents • Evaluation Frameworks • Prompt & Context Engineering • RAG & Grounding • Python • TypeScript • FastAPI / Docker

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present | USA

Jr. AI/ML Engineer Trainee - Healthcare AI, Claude / LLM Integration, Production Delivery

- Built **RAG systems** for clinical knowledge retrieval and healthcare documentation search; recursive semantic chunking + transformer embeddings improved contextual retrieval precision **~35%** and reduced irrelevant retrieval **>30%**. Dedicated **eval harness** raised grounded response alignment above **90%**.
- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent response stability **~25%**.
- Shipped predictive ML pipelines (scikit-learn, XGBoost); improved recall **15-20%** on high-risk patient categories while holding precision **>90%**.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging; reduced post-deployment defects **~30%**.
- Maintained HIPAA-conscious data governance: de-identification, data lineage, stakeholder-facing system-limitation docs - the documentation discipline mission-driven partners depend on.

Suvidha Foundation (Nonprofit)

Jan 2022 - Mar 2022 | India

Machine Learning Engineer - Nonprofit Applied AI, Document Q&A RAG

- Built a **transformer-based video summarization and highlight-generation system** across **5,000+ recorded sessions**; transcript preprocessing + hierarchical summarization for long-context + timestamp-aligned clip extraction reduced manual review time **60-70%** (hours -> under 5 minutes per video).
- Achieved **~85% alignment** between AI-selected highlights and human-curated key moments.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs **~30%** and raised contextual answer accuracy **>85%**.
- Deployed the stack as a **Flask** API with a lightweight web interface for non-technical staff - building for mission-driven users is its own discipline.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023 | India

Database & DevOps Performance Engineer (Intern) - Compliance-Sensitive Enterprise

- Built **CI/CD pipelines (Jenkins)** for SQL and stored-procedure deploys with release validation + rollback checkpoints; reduced deployment errors **>30%**, improved release cycle **~35-40%**.
- Tuned T-SQL stored procedures and reconciliation queries; cut query execution time **~20%**; supported **RBAC + audit logging** for financial modules in a compliance-sensitive oil & gas environment.

PROJECTS

Manga Lens - Chrome Extension with Claude Sonnet Integration (shipped, Chrome Web Store)

Full-stack browser extension built independently. Integrates **Claude Sonnet** alongside GPT-4o mini, GPT-4.1 Nano, and Ollama/minicpm-v local behind one interface with provider-specific payload handling. Manifest V3, content scripts, service workers, seven-day translation cache, per-domain selector configs for 29 sites, privacy policy, narrowed host permissions.

Production Claude API integration in shipped user-facing software.

Claude Sonnet, multi-provider AI, TypeScript, Manifest V3

Agentic Healthcare Claims Pipeline - MCP-Adjacent Multi-Agent System

Multi-agent pipeline (Intake -> Validation -> Consistency -> Duplicate Detection -> Fraud Risk) with **schema-validated JSON contracts between agents** - the same tool-discipline pattern MCP server design depends on. RAG-grounded CPT/ICD validation, ANN duplicate detection, explainable risk scoring with **audit-ready reasoning traces**.

Multi-agent, JSON schema contracts (MCP-adjacent), RAG, vector search, audit traces

Dream Decoder - Multimodal Creative Intelligence Platform

Full-stack **FastAPI + React/TypeScript/Vite/Tailwind** app coordinating multimodal APIs (Perplexity Sonar + Replicate).

Intermediate structured prompt transformation layers improved contextual alignment ~**30%** and first-pass image success ~**25-30%**.

FastAPI, TypeScript, structured prompt orchestration

E-Farming Digital Marketplace (Founder & Full-Stack Developer)

Solo founder build of a PHP/MySQL/Bootstrap AgriTech marketplace connecting small-scale farmers with buyers. Onboarded **80-120 active users** in phase one. Cart, reviews, and community blogging features drove repeat engagement.

PHP, MySQL, Bootstrap, founder, mission-driven

EDUCATION

Master's Degree, Computer Science - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages & Frameworks: Python, TypeScript, FastAPI, Flask, SQL (PostgreSQL, T-SQL, SQLite), React, Node.js

LLM & GenAI: Claude API, MCP-adjacent multi-agent design, agent skills, sub-agents, prompt engineering, context engineering, evaluation frameworks, RAG, grounding, embeddings, vector search

Infra & DevOps: Docker, Jenkins, CI/CD, RESTful APIs, observability/logging

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex

Mission-Driven / Compliance: Nonprofit AI delivery, HIPAA-conscious governance, RBAC, audit logging, building for non-technical users